Beyond Semantic Prosody: Reassessing the Directionality of Connotative Meaning in Lexical Semantics

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Abstract

The theory of semantic prosody posits that words acquire evaluative coloring through repeated collocation with positively or negatively charged contexts. While this usage-based framework effectively describes distributional patterns, it faces persistent theoretical challenges regarding directionality: does usage shape meaning, or does meaning shape usage? This paper demonstrates that for a significant class of frequently studied words, inherent lexical-semantic profiles act as primary filters that attract words to congruent discourse contexts, with collocational patterns emerging as consequences rather than causes of connotative meaning. Through systematic analysis of *cause*, *utterly*, *commit*, *set in*, *foster*, and *regime*, integrating published corpus analyses with theoretical linguistics, this study shows that their attested collocational skews reflect underlying semantic features—agentive, intensive, volitionally binding, gradual-onset, facilitative, and system-control profiles, respectively. Counterevidence from neutral and positive contexts in multiple corpora undermines claims of fixed evaluative prosodies while supporting profile-based explanations. Diachronic evidence demonstrates that semantic profiles remain stable across centuries even as specific collocational patterns vary, supporting the claim that profiles drive usage rather than absorbing evaluative charge from it. This reconceptualization addresses the problematic lexical-encyclopedic dichotomy, demonstrating through evidence from large language models, language testing, and cross-linguistic research that the traditional boundary between lexical meaning and world knowledge is far more permeable than conventional accounts acknowledge. The findings have practical implications for lexicography, natural language processing, and language pedagogy.

Keywords: semantic prosody, lexical semantics, collocation, corpus linguistics, connotation, semantic profiles, directionality, usage-based linguistics, language models, lexical-encyclopedic distinction

Beyond Semantic Prosody: Reassessing the Directionality of Connotative Meaning in Lexical Semantics

The advent of large-scale corpus linguistics has fundamentally transformed our understanding of lexical behavior, revealing patterns of word usage that remain largely inaccessible to introspection. Among the most theoretically significant discoveries is the consistent statistical patterning of words within evaluatively marked contexts—regularities so robust that they demand explanation yet so subtle that native speakers rarely consciously perceive them. The verb *cause* predominantly collocates with negative outcomes such as *damage*, *death*, and *problems*; the adverb *utterly* intensifies predominantly negative adjectives; and nouns like *regime* appear disproportionately alongside modifiers such as *military*, *communist*, and *fascist* (Louw, 1993; Sinclair, 1991; Stubbs, 2001). These patterns reveal what Stubbs (2001, p. 65) characterizes as "a level of organization beyond the reach of intuition," one that is "neither random nor wholly predictable from syntax or semantics alone."

The dominant theoretical framework addressing these patterns has been the theory of semantic prosody, pioneered by Sinclair (1991) and Louw (1993). In its strong form, this usage-based model posits that words gradually absorb or are imbued with evaluative coloring through repeated co-occurrence with charged collocates. The explanatory direction flows from context to word: *cause* acquires negative associations because it frequently appears with negative direct objects. While descriptively powerful and empirically grounded, this model represents what might be characterized as an explanatory shortcut—it documents correlation without adequately addressing causality, struggles to account for systematic counterexamples, and leaves unresolved the theoretical status of patterns that are statistically robust yet apparently psychologically non-salient.

The recent success of large language models trained purely on text statistics makes this theoretical question increasingly urgent. Systems like GPT-4 and Claude demonstrate striking semantic competence despite learning solely from distributional patterns in massive text corpora. They appropriately use *cause* with both negative outcomes and expressions like *cause for celebration*, distinguish when *foster* applies to negative developments like *resentment*, and deploy *regime* felicitously across political and non-political contexts. This competence appears puzzling under strong semantic prosody theory: if connotative meaning were truly absorbed from collocational frequency alone, models should overgeneralize the dominant pattern and resist counterexamples. Yet they do not. Their success suggests that corpus patterns may encode something more structured than simple evaluative associations—something that permits principled generalization beyond statistical frequencies, while also revealing a structural ceiling on what statistical learning of any scale can achieve.

This paper demonstrates a reconceptualization that integrates insights from both usage-based and formal semantic traditions. Rather than treating directionality as strictly unidirectional, it shows that for many canonical semantic prosody cases, inherent lexical-semantic profiles—structured sets of semantic features defining the prototypical nature of events, entities, or properties denoted by a lexical item—act as primary filters that attract words to congruent discourse contexts. Collocational patterns, rather than being constitutive of connotative meaning, emerge as pragmatic reflexes of the interaction between stable word meanings and the distributional asymmetries inherent in human discourse priorities. Through detailed reanalysis of six extensively studied cases (*cause*, *utterly*, *commit*, *set in*, *foster*, *regime*), this study demonstrates that their observed distributional skews are better explained as consequences of profiles involving agentivity, intensity, volitional binding, gradual onset, facilitation, and system-control, respectively, rather than as absorbed evaluative features. Diachronic evidence from historical corpora and the Oxford English Dictionary shows that these semantic profiles have remained remarkably stable across centuries of English usage, even as specific collocational patterns have shifted with changing discourse priorities—a finding that directly supports the profile-based account over absorption models.

The methodological framework employed in this study addresses the most pressing concern such reanalyses face: the appearance of circularity. If profiles are inferred from the same distributional patterns they are meant to explain, the explanatory loop is closed before it begins. The present account avoids this by grounding profile characterizations in lexicographic definitions—particularly the historical entries of the Oxford English Dictionary—which constitute an evidential source independent of the synchronic corpus data that semantic prosody research has examined. Profiles thus serve as theoretical commitments derived from one body of evidence, generating predictions tested against another. A further empirical prediction—that profile-driven distributions should be scale-invariant across corpus sizes—is articulated explicitly and described in testable form, with the test deferred to future work when computational means become available. The prediction stands as a theoretical commitment regardless of when it is run.

This reconceptualization engages a fundamental theoretical question that has constrained semantic research: the boundary between lexical meaning and encyclopedic knowledge. The semantic profiles under investigation—features like agentivity, facilitation, and system-control—inherently span what has traditionally been treated as the lexical-encyclopedic divide. The present paper does not develop this consequence into a full theory; that has been done in separate work (Mozahid, 2025, 2026a, 2026b). It does, however, register the consequence: the directionality thesis defended here is not neutral with respect to the lex-enc boundary but actively dissolves the categorical version of it.

This does not entail complete dissolution of all distinctions between linguistic and factual knowledge. Rather, it demonstrates that the boundary is gradient and context-dependent rather than categorical and fixed. By moving from a model of *connotative absorption* toward one emphasizing *profile-constrained deployment*, this framework offers solutions to persistent theoretical problems while providing actionable insights for understanding both human linguistic competence and the emerging capabilities of statistical language models.

Theoretical Background: Semantic Prosody and the Directionality Debate

The concept of semantic prosody emerged from corpus-driven approaches to meaning in the late 1980s and early 1990s, representing a significant shift in how linguists conceptualized connotation. Louw (1993, p. 157) defined it as "a consistent aura of meaning with which a form is imbued by its collocates," while Sinclair (1991, p. 112) described it as the "reason why a speaker chooses a particular word" based on its "attitudinal function." The methodology appeared straightforward: through systematic analysis of large text corpora, researchers could identify statistically significant collocational patterns that revealed words' hidden evaluative charges. Early canonical examples included *set in* with undesirable states (*rot*, *decay*), *cause* with negative outcomes, and *commit* with transgressions (Louw, 1993; Sinclair, 1991).

The theoretical interpretation accompanying these empirical findings aligned with broader usage-based and constructional approaches to language, particularly those emphasizing frequency effects and statistical learning in shaping linguistic representation (Bybee, 2006, 2010; Goldberg, 2006). From this perspective, meaning is fundamentally context-dependent and continuously shaped by patterns of use, with lexical entries conceived as permeable and constantly updated by encountered statistical regularities. Semantic prosody, in this view, represents an emergent property of word-in-use, where corpus distributions constitute a formative force in the development of connotative meaning. As Hoey (2005, p. 13) articulates, "every word is primed for use through our encounters with it," and this priming includes not just grammatical and semantic information but also pragmatic and evaluative associations.

This original framework has not stood still. Subsequent prosody scholarship has retreated from the strongest form of the absorption thesis. Partington (2004) introduced the distinction between *semantic prosody* and *semantic preference*, separating evaluative orientation from the broader phenomenon of a word's tendency to co-occur with semantically related items. Hunston (2007) explicitly cautioned that "corpus data are evidence of meaning rather than meaning itself," a methodological reservation that already concedes much of the directionality question. The position the present paper opposes is therefore not a position currently held in full strength by any major contemporary prosody scholar. What it opposes is the residual commitment, still present across the field's vocabulary and explanatory habits, that distributional skews constitute the connotative meaning rather than diagnose it. Even the more nuanced contemporary positions retain the assumption that the locus of explanation is the corpus pattern itself rather than the lexical-semantic architecture that generates that pattern.

This persistent commitment encounters the directionality problem in its sharpest form: does usage shape meaning, or does meaning shape usage? Stubbs (2001, p. 105) acknowledges what he terms "the introspection problem"—the robust collocational patterns revealed by corpora frequently surprise native speakers who intuitively regard words like *cause* as evaluatively neutral. This disconnect between corpus-revealed patterns and speaker intuition raises questions about the psychological reality and theoretical status of semantic prosodies. More critically, while correlation between a word and evaluative contexts may be empirically clear, establishing causal direction remains theoretically problematic.

Several scholars have identified limitations in the strong prosodic account. Whitsitt (2005) critiques the tendency to reify corpus patterns into word meanings, arguing that semantic prosody conflates "meaning proper" with "meaning effects" arising from pragmatic inference. Stewart (2010) demonstrates through psycholinguistic experiments that purported prosodies may not be cognitively represented as stable features but rather emerge through online processing. More recently, Blumenthal-Dramé (2016) has shown using behavioral data that collocational strength varies considerably across speakers and correlates with individual exposure patterns, suggesting that semantic prosody may be less a property of words in the language system than a reflection of personal linguistic experience.

An alternative hypothesis, rooted in formal and cognitive semantic traditions, posits that words carry encoded semantic features and structured representations that constrain their appropriate deployment (Cruse, 1986; Pustejovsky, 1995). From this perspective, collocational patterns represent outputs of a filtering process whereby lexical semantic profiles make words natural fits for particular conceptual contexts. The critical weakness of strong absorption models becomes apparent when considering systematic exceptions: if negative prosody were purely a product of frequency-based association, counterexamples should sound anomalous or require special pragmatic licensing. Yet as documented below, such counterexamples exist, are comprehensible, and often derive their rhetorical effects from tension with core semantic profiles rather than absorbed prosodies.

The success of large language models in acquiring human-like semantic competence provides an additional perspective on this debate. These systems learn exclusively from distributional patterns in text corpora without access to extralinguistic grounding or explicit semantic instruction (Brown et al., 2020; OpenAI, 2023; Anthropic, 2024). If semantic prosodies were properties that words absorbed through repeated contextual exposure in a way fundamentally divorced from underlying semantic structure, we would expect statistical models to learn rigid collocational preferences that resist generalization. Instead, state-of-the-art language models demonstrate flexible deployment of words across diverse contexts, including systematic counterexamples to dominant distributional patterns. This suggests that the statistical regularities in corpora may encode information not merely about surface co-occurrence but about deeper semantic constraints—precisely the kind of structured profiles this paper argues are primary.

Alternative Theoretical Accounts

Before proceeding to the profile-based framework advocated here, it is essential to consider why other theoretical approaches fail to adequately explain the observed patterns. Four alternative accounts merit systematic evaluation.

First, a pure frequency or statistical association account might claim that collocational patterns emerge from simple co-occurrence statistics without invoking semantic structure. Words appear together because they have appeared together in the past, creating self-reinforcing statistical tendencies. This approach fails on multiple grounds. It cannot explain why certain counterexamples to dominant patterns sound perfectly natural while others sound marked. If *cause* developed negative associations purely through frequency, then *cause celebration* should sound as anomalous as *cause for sadness* sounds natural—yet the former is comprehensible while the latter sounds redundant. Pure frequency accounts also cannot explain the systematic generalization exhibited by language models trained on limited data, which appropriately extend words to novel contexts they may never have encountered.

Second, a Construction Grammar approach (Goldberg, 2006) might argue that evaluative meanings emerge from larger constructional patterns rather than individual lexical items. While Construction Grammar provides valuable insights into how meaning emerges from form-meaning pairings at various levels of abstraction, it does not eliminate the need for lexical semantic analysis. The patterns documented for *cause*, *utterly*, and other words persist across multiple constructions and syntactic frames. Moreover, Construction Grammar itself requires specification of which lexical items can fill particular slots in constructions—precisely the kind of constraint that semantic profiles provide. The two approaches are complementary rather than competing, with profiles explaining word-level constraints that interact with constructional meanings.

Third, a conventional implicature account might claim that evaluative associations arise through Gricean mechanisms, where repeated pragmatic inferences become conventionalized over time (Traugott & Dasher, 2002). While diachronic conventionalization certainly occurs, this account makes incorrect predictions about cancelability. Conventional implicatures are typically cancelable without contradiction, but alleged semantic prosodies show different behavior. One can felicitously say *cause celebration* without any sense of canceling an implicature—the expression simply describes direct causation of a positive outcome. If the negativity were a conventional implicature, explicit cancellation should be necessary, but it is not.

Fourth, a pure pragmatic account might argue that distributional patterns reflect only discourse preferences without any connection to lexical semantics. Speakers simply choose to talk more about negative causation, negative intensity, and authoritarian regimes, creating statistical skews that have no semantic reality. This account fails to explain the systematic nature of the patterns and their stability across diverse discourse contexts. It also predicts far more variation across genres and registers than actually occurs—the agentive profile of *cause* attracts it to forceful impositions in academic writing, journalism, and casual conversation alike.

The profile-based framework advocated here avoids these problems by proposing that lexical entries encode semantic profiles—sets of features defining prototypical characteristics of denoted events, entities, or properties—which act as constraints on felicitous usage. Collocational patterns revealed in corpora represent the joint output of these constraints interacting with discourse pragmatics and the asymmetric salience of different event types in human communication. What semantic prosody theory identifies as absorbed evaluation is better characterized as a diagnostic of underlying semantic architecture.

Analytical Framework for Case Studies

The following case studies apply a consistent analytical framework with four components. First, each analysis documents the established corpus-based distributional pattern that has been interpreted as evidence for semantic prosody. Second, an alternative semantic profile is proposed based on established semantic features from formal and cognitive semantic frameworks, with profile characterization grounded in lexicographic evidence independent of the corpus patterns being explained. Third, the analysis demonstrates how this profile predicts the observed collocational pattern through interaction with discourse pragmatics and conceptual structure. Fourth, counterevidence is examined—specifically, documented cases where the word appears in neutral or positively valenced contexts that would be anomalous under a strong prosodic account but are fully licensed by the proposed semantic profile.

This framework demonstrates that apparent "prosodies" reflect stable semantic profiles interacting with discourse pragmatics rather than absorbed evaluative meanings. The critical theoretical move is showing that counterexamples, while statistically less frequent, are fully comprehensible without pragmatic strain, whereas they would constitute anomalies requiring special explanation under a strong prosodic account. Additionally, the behavior of large language models provides converging evidence: these systems, trained purely on distributional patterns, demonstrate the same profile-based flexibility in deploying words across both prototypical and counterexample contexts. This generalization capacity would be mysterious if corpus patterns encoded only surface evaluative charges, but it becomes explicable if those patterns carry information about the underlying semantic architecture that licenses both typical and atypical uses.

Methodological Approach

This study employs a theoretically driven reanalysis of existing published corpus findings rather than conducting independent large-scale corpus queries. This approach is justified on several grounds. First, the empirical patterns under examination—the collocational distributions of *cause*, *utterly*, *commit*, *set in*, *foster*, and *regime*—are well-established and extensively documented in the semantic prosody literature (Louw, 1993; Sinclair, 1991; Stubbs, 2001; Partington, 2004). These patterns are not in empirical dispute; rather, the theoretical question concerns their proper explanation. The corpus-based distributional facts have been replicated across multiple large corpora including the Bank of English, the British National Corpus, and the Corpus of Contemporary American English, with consistent findings regarding collocational tendencies.

Second, the focus here is explicitly on theoretical reinterpretation rather than pattern discovery—demonstrating that existing corpus evidence, when viewed through the lens of lexical semantic profiles, supports an alternative causal model that better accounts for both the dominant patterns and the systematic exceptions. The contribution is therefore primarily theoretical: showing that semantic profile-based explanations offer a more parsimonious and comprehensive account of patterns that semantic prosody theory has identified but perhaps mislabeled. This type of theoretical work does not require new empirical data collection when the relevant patterns are already well-documented and accepted in the literature.

Third, theoretical reanalysis represents a legitimate and valuable form of scientific contribution. The history of linguistics is replete with theoretical advances that emerged not from new data but from reconceptualizing the significance of known phenomena—Chomsky's poverty of stimulus argument, for instance, reinterpreted rather than discovered the facts of language acquisition.

The Independence of Profile Characterization from Corpus Distribution

A potential objection to the analytical framework outlined above concerns circularity. If semantic profiles are inferred from the same distributional patterns they are meant to explain, the explanatory loop is closed before it begins: the profile is constructed from observations of the corpus, and the corpus is then explained by appeal to the profile. The objection is serious, and the framework deployed here is structured specifically to avoid it.

Profile characterizations in this study are not derived from the synchronic corpus distributions documented by Stubbs (2001) or Partington (2004). They are derived from lexicographic evidence—principally the historical entries of the Oxford English Dictionary—supplemented by core meaning specifications in established lexical-semantic frameworks (Cruse, 1986; Pustejovsky, 1995; Talmy, 1988, 2000; Dowty, 1991). This is an evidential source independent of the corpus data under examination, and its independence is methodologically essential.

The OED entries for the words analyzed here record citation evidence spanning many centuries, with the earliest attestations for *cause* drawn from fourteenth-century texts, for *commit* and *foster* from late Middle English, and for *regime* from late eighteenth-century usage. The core meaning specifications established by these historical records were determined through lexicographic practice that predates modern corpus linguistics entirely. Whatever profiles the OED encodes for these words were established before Sinclair, Louw, Stubbs, and Partington had compiled the synchronic distributional patterns that semantic prosody theory takes as its evidence. The directionality of evidential dependence is therefore not closed but open: profiles characterized from one source generate predictions tested against another.

A skeptic might press that lexicographic definitions are themselves partly based on usage evidence, so the independence is not absolute. The response is straightforward. The OED's earliest citations for these words predate the corpora used by contemporary semantic prosody researchers by several centuries. The core meanings established from Middle English or Early Modern English sources cannot be circularly derived from twenty-first-century Bank of English frequency counts. The independence is genuine, even if not metaphysically pure. What matters methodologically is that the profile characterization and the distributional pattern being explained come from non-overlapping bodies of evidence, and they do.

This structural feature of the analysis—deriving profiles from one evidential source and testing them against another—is what enables corpus distributions to function as confirmatory evidence rather than as both source and explanandum. The reanalysis presented in the case studies should be understood in these terms: profiles characterized from independent lexicographic and theoretical sources are used to predict, and to be tested against, the distributional patterns documented in the corpus linguistic literature.

A Testable Prediction: The Scale-Invariance of Profile-Driven Distributions

The framework defended here generates a further empirical prediction that distinguishes it from accounts treating meaning as constructed from accumulated distributional exposure. If semantic profiles are primary and drive the distributional patterns observed in corpora, then the distributional signature of a given profile should be detectable at multiple corpus scales. A small corpus and a large corpus should reveal the same profile-driven structure, because the structure is in the profile itself, not in the quantity of distributional data sampled. The corpus is sampling a pre-existing semantic constraint, not constructing it through accumulation.

The contrast with an absorption account is sharp. If meaning were genuinely constructed from accumulated distributional exposure, profile-relevant distributional signatures should be unstable at small corpus sizes and only consolidate as data accumulates. Different small corpora drawn from the same population should yield divergent profile characterizations, with convergence emerging only at scale. Under the profile-based account, by contrast, the relative rankings and qualitative shape of profile-relevant collocational tendencies should be largely scale-invariant. Larger corpora should sharpen and stabilize the signal, but they should not introduce qualitatively new structure absent at smaller scales.

The test is straightforward in principle. Take a freely available large corpus—the British National Corpus, the openly accessible subset of the Corpus of Contemporary American English, the Open American National Corpus, or processed Wikipedia dumps. Partition it into subsets at progressively decreasing scales—halves, quarters, eighths, sixteenths. For a defined target set of items including the six examined in this study and additional comparison cases, compute profile-relevant collocational distributions at each scale. Compare the rankings and qualitative shape of distributions across scales. The profile-based account predicts substantial convergence of rankings and shape even at modest scales. An absorption account predicts substantial reorganization of structure as scale increases.

A nuance worth registering at the outset is that the prediction should hold most strongly for high-fixity profile features (those corresponding to what the related work characterizes as high alpha) and should hold more weakly for low-fixity features. Profile components that are genuinely fixed in the lexicon should stabilize at small corpus sizes; profile components that are more variable across speakers and registers should remain variable at all scales. This expected gradient pattern is itself predicted by the framework and would not be predicted by an absorption account, which has no principled basis for distinguishing among profile components in this way.

The test is fully testable using freely available corpora and standard concordance tools (AntConc, the Sketch Engine free tier, Python with NLTK), and it does not require institutional resources beyond what an individual researcher with basic computational tools can deploy. It has not been run as part of the present study. The prediction is offered here as a theoretical commitment of the profile-based account, to be confirmed or disconfirmed by empirical work as means become available. The forecast stands on its own as a theoretical commitment regardless of the timing of its empirical resolution. A framework that generates falsifiable predictions before it can run them is doing the work theoretical frameworks are supposed to do; one that withholds predictions until they can be operationally confirmed has weakened itself unnecessarily.

It is important to acknowledge further methodological limitations. This study does not provide original quantitative data on frequency distributions, statistical significance tests, or systematic sampling procedures. Furthermore, this study does not directly test psychological reality through experimental methods such as priming studies or acceptability judgments, though it draws on existing psycholinguistic research where relevant (e.g., Stewart, 2010; Blumenthal-Dramé, 2016). The final section proposes specific experimental designs that could directly test the predictions of the profile-based account, providing a roadmap for future empirical work that would complement the theoretical analysis presented here.

Case Studies

Case 1: Cause—Agentive Profile, Not Negative Prosody

The English verb *cause* represents perhaps the most frequently cited example in semantic prosody literature. Corpus studies have uniformly documented its strong statistical association with nouns denoting adverse outcomes. Stubbs (2001, pp. 106–107) reports that in the Bank of English corpus, *cause* collocates most frequently with words like *problems*, *damage*, *concern*, *trouble*, and *death*. Sinclair (1991) estimates that negative collocates account for approximately 70% of transitive uses. The standard prosodic interpretation is that through repeated co-occurrence with negative outcomes, the neutral causative verb has absorbed a negative evaluative charge, becoming semantically tinged with adversity.

An alternative explanation emerges from the verb's lexical-semantic profile as characterized by the OED and by theoretical work on causation predating contemporary corpus linguistics. The profile encodes direct, forceful agentivity. Unlike more neutral causative expressions such as *lead to* or *result in*, *cause* specifies scenarios where a salient agent or potent force directly and effectively brings about a change of state. This profile aligns with Dowty's (1991) proto-agent properties, particularly the features of volitional involvement and causing change in another participant. The semantic core of *cause* involves an imposition of force that overcomes resistance to produce an effect. This is evident in entailment patterns: *X brought about Y* readily entails *X caused Y*, but *X resulted in Y* does not straightforwardly entail *X caused Y* without implying a more direct, forceful causal link.

Given this agentive profile, the observed distributional skew becomes pragmatically predictable. In human discourse, the most salient and reportable negative events—harm, injury, death, destruction, suffering—are prototypically conceptualized as states actively imposed by external forces. Narratives of adversity naturally invoke force-dynamic schemas (Talmy, 1988, 2000) where an agonist overcomes an antagonist. The agentive semantics of *cause* provides an optimal linguistic match for this conceptual structure. Conversely, many positive outcomes—happiness, prosperity, understanding, growth—are less frequently framed as direct objects of forceful agentive imposition. Such states are more commonly described as emerging, developing, being fostered, or arising naturally, hence the preference for verbs like *foster*, *promote*, or *facilitate* in positive contexts.

Critically, corpus evidence reveals systematic counterexamples that undermine a fixed negative prosody while supporting the agentive profile account. While statistically less frequent, *cause* appears with neutral or positive direct objects. COCA (Davies, 2008–) yields examples including *cause a sensation*, *cause excitement*, *cause great joy*, *cause celebration*, and *cause a stir*. The phrase *cause for celebration* is itself a common, lexicalized positive construction. These uses are not anomalous or ironically marked; they describe situations where an agent is perceived as directly and forcefully bringing about a positive or neutral change. A political reform that *causes excitement among voters* or a discovery that *causes celebration in the scientific community* invokes the same agentive semantics as negative cases—the difference lies in the evaluation of the outcome, not in the verb's meaning.

The distributional skew is therefore more accurately characterized as agentive rather than negative. The negativity represents a pragmatic reflex of discourse priorities—we more frequently need to identify and report forceful agents of harm than forceful agents of good—rather than a semantic feature encoded in the verb itself. This reanalysis resolves Stubbs's (2001) introspection problem: speakers correctly intuit *cause* as evaluatively neutral because agentivity, not negativity, constitutes its core meaning. The negative skew is an observable consequence of how this neutral semantic profile interacts with asymmetries in what we talk about.

Large language models trained on corpus data demonstrate this same pattern of flexible deployment. They readily generate both *policies that cause harm* and *discoveries that cause excitement* because the distributional evidence, while skewed toward negative contexts, carries information about agentive force-dynamic structure rather than negativity per se. The semantic profile leaves statistical traces that survive the transformation from meaning to usage, enabling models to extract the abstract constraint rather than memorizing rigid associations.

Case 2: Utterly—Intensive Profile, Not Negative Prosody

The adverb *utterly* constitutes another canonical case in the semantic prosody literature. Louw (1993) and Sinclair (1991) document its predominant function as an intensifier for negative adjectives: *utterly disastrous*, *utterly appalling*, *utterly ridiculous*. Partington (2004, p. 145) quantifies this pattern in the Cambridge International Corpus, finding that 78% of *utterly*'s collocates are negatively valenced adjectives. The prosodic conclusion drawn is that *utterly* has absorbed a negative semantic prosody, functioning specifically as a magnifier for undesirable qualities.

However, examination of *utterly*'s semantic profile reveals an alternative account. Its core meaning denotes completeness, totality, and extremity—*to an absolute or extreme degree*. This intensive profile is its defining lexical feature, deriving etymologically from *utter* meaning *complete* or *absolute* (Oxford English Dictionary, 2023), with attestations stretching back through Early Modern and Middle English. The critical question becomes: why does an intensifier meaning *completely* show distributional preference for negative adjectives? The answer lies in discourse pragmatics.

First, there exists greater pragmatic pressure to intensify negative evaluations in communication. Complaints, warnings, and expressions of disapproval deploy extreme language to signal seriousness and justify subsequent action. As Pomerantz (1984) demonstrates in conversation analysis, second assessments that upgrade negative evaluations are more common and less face-threatening than those that upgrade positive ones. Second, intensifying positive qualities with *utterly* can risk hyperbole or sound disproportionately effusive. *Utterly wonderful* may be perceived as more exaggerated than *utterly terrible*, not because the adverb carries negativity, but because extreme intensification of praise requires more contextual support to avoid sounding insincere.

Corpus evidence provides crucial counterexamples. Searches in COCA and BNC yield numerous instances of *utterly* modifying positive or neutral adjectives: *utterly beautiful*, *utterly compelling*, *utterly delightful*, *utterly charming*, *utterly convincing*, and *utterly captivating*

(Davies, 2008–; BNC Consortium, 2007). If *utterly* possessed an encoded negative prosody, positive collocations should sound anomalous or require ironic interpretation. Yet in context they function straightforwardly: *her performance was utterly compelling* intensifies the positive assessment without semantic tension. The adverb operates precisely as its intensive profile predicts—it magnifies the degree of the adjective's denoted property, regardless of evaluative polarity. The distributional skew toward negativity is better characterized as a pragmatic tendency emerging from interaction between the intensive semantic feature and conventions of evaluative discourse.

Case 3: Commit—Volitional-Binding Profile, Not Negative Prosody

The verb *commit* frequently appears in discussions of semantic prosody due to its high-frequency collocates: *commit a crime*, *commit murder*, *commit suicide*, *commit fraud*. Stubbs (2001, p. 109) reports that in the Bank of English corpus, over 60% of transitive uses of *commit* involve direct objects denoting illicit, harmful, or transgressive acts. The prosodic interpretation posits that through association with transgressions, the verb itself has become negatively charged.

Analysis of *commit*'s semantic core, as recorded in the OED's historical entries and theorized by Talmy (2000), reveals not negativity but volitional and binding engagement. To commit is to dedicate, pledge, or bind oneself or something to a particular course, purpose, or place. This profile involves several interlocking features: conscious volition (the act is deliberate), consequentiality (the commitment has weight), and entrenchment (the binding is not easily reversed). These features align with what Talmy (2000) analyzes as force dynamics involving internal self-compulsion and external constraint. The verb specifies not just that an action occurred but that an agent consciously bound themselves to that action's execution or consequences.

This volitional-binding profile elegantly predicts the observed collocational pattern. The acts most notoriously accompanying *commit*—crimes, sins, transgressions—are precisely those conceptualized as representing profound, binding, conscious choices to violate norms. The gravity of the act matches the gravity implied by *commit*'s sense of binding engagement. Crucially, this same profile fully licenses *commit*'s extensive use in neutral and positive contexts. We routinely encounter *commit to a relationship*, *commit resources to a project*, *commit oneself to a goal*, *commit funds to research*, or *commit troops to battle*. In legal and institutional registers, one *commits a patient to psychiatric care* or *commits a defendant to trial*. These uses are not marginal but central to the verb's polysemy. All involve deliberate, binding allocation or dedication. The negative skew emerges because binding oneself to a norm-violating act represents a highly salient and reportable event type in social discourse.

Case 4: Set In—Gradual-Onset Profile, Not Negative Prosody

The phrasal verb *set in* represents a classic case in semantic prosody studies, famous for collocating with undesirable states: *rot set in*, *decay set in*, *despair set in*, *boredom set in*. Louw (1993, p. 163) identifies it as exhibiting "unfailingly negative" prosody. The interpretation follows the standard pattern: through habitual association with negative states, the phrase has absorbed negative coloring such that any state which sets in is implicitly construed as undesirable.

Analysis of *set in*'s inherent aspectual semantics suggests an alternative account. The phrase encodes the gradual, persistent, and often irreversible onset or establishment of a state or process. It combines the inchoative meaning of *begin* with implications of entrenchment and duration. This aspectual profile aligns with Vendler's (1957) category of achievements transitioning into states, but with specific additional features: the onset is not punctual but gradual, the resulting state is persistent, and reversal is difficult.

This gradual-onset profile naturally aligns with a class of undesirable states because, in human experience, many negative conditions are characterized by insidious, creeping, hard-to-reverse progression. Physical decay, emotional malaise, social decline, and environmental degradation all exemplify processes we observe and lament as they gradually establish themselves. Conversely, positive states are less frequently conceptualized through this aspectual frame. We more often describe positive developments as achievements or rapid improvements rather than as slow, creeping processes.

Nevertheless, counterexamples exist and are semantically coherent when the aspectual profile is appropriate. Documented examples include *a sense of peace finally set in*, *as the medication took effect, healing set in*, *a comfortable routine set in*, and *after the chaos, calm set in* (Davies, 2008–). These uses leverage the same gradual-onset semantics: peace, healing, routine, and calm can all be conceptualized as states that gradually establish themselves and persist. The prosody of *set in* is therefore more accurately described as gradual-onset or persistent establishment, with the negativity representing a strong pragmatic association rather than a fixed semantic feature.

Case 5: Foster—Facilitative Profile, Not Positive Prosody

The verb *foster* offers a valuable contrast case, often presented as exhibiting positive semantic prosody. Stubbs (2001, p. 110) documents its frequent collocation with desirable outcomes: *foster growth*, *foster development*, *foster understanding*, *foster cooperation*. Statistical analysis in the Bank of English shows that approximately 80% of *foster*'s direct objects are positively valenced.

However, *foster*'s lexical-semantic profile provides a superior explanation. The verb encodes facilitative, nurturing, or enabling causation. It depicts scenarios where an agent or environment provides conditions conducive to the development or emergence of something, typically over time and through gradual process. This facilitative profile contrasts systematically with the forceful, direct agentivity of *cause*. Where *cause* implies imposition, *foster* implies support; where *cause* is direct, *foster* is indirect; where *cause* is immediate, *foster* is gradual.

This facilitative meaning naturally aligns with contexts where positive outcomes are cultivated: child development, education, economic growth, social harmony, intergroup understanding. These represent domains where gentle encouragement and supportive conditions are conceptually appropriate. Importantly, this analysis predicts that *foster* should be felicitous in any context involving gradual enablement, regardless of ultimate evaluation.

Corpus evidence confirms this prediction, revealing that to claim *foster* has a positive prosody is an overgeneralization. The verb's facilitative profile is evaluatively neutral and can apply to negative developments. Documented examples include *foster resentment*, *foster dependence*, *foster inequality*, *foster corruption*, *foster distrust*, *foster illusions*, and *foster complacency* (Davies, 2008–). These are not ironic or marked uses; they straightforwardly describe the nurturing of conditions leading to negative outcomes. *Policies that foster inequality* means policies that create conditions enabling inequality to develop. The verb's meaning—to enable gradual development—remains constant; what varies is the evaluation of what is being developed. This conclusively demonstrates that positivity is not a semantic feature of *foster* but a frequent pragmatic association.

Case 6: Regime—System-Control Profile, Not Negative Prosody

The noun *regime* serves as a powerful illustration of the introspection problem. Channell (2000, p. 46) discovered that the most frequent collocates of *regime* in corpus data are words like *military*, *communist*, *ancien*, *Nazi*, *Soviet*, *Vichy*, *fascist*, *present*, and *Iraqi*. The standard prosodic account would interpret this as evidence that *regime* has absorbed a negative semantic prosody, becoming a pejorative term for a government.

An analysis based on semantic profile reveals a different causality. The core meaning of *regime*, established in late eighteenth-century English borrowings from French *régime* and recorded through its OED entries, involves systematic, structured, and often imposed control or administration. Its semantic profile is one of system-control, characterized by formality, systemicity, and top-down authority. This profile is evaluatively neutral but naturally aligns with contexts describing organized systems of rule, particularly those that are formalized, rigid, or non-democratic.

The striking negative skew emerges predictably from this interaction. In political and historical discourse, particularly in Western media and academic analysis, systems of control that are imposed, authoritarian, or totalitarian are frequent topics of discussion and are overwhelmingly evaluated negatively. Therefore, the word *regime* is naturally drawn to these contexts because its meaning matches the conceptual structure of what is being described. The negativity resides in the evaluation of the referent, not in the semantic entry for the word.

Crucially, counterexamples in neutral or positive contexts are fully licensed by the same core meaning. Corpus data and general usage readily provide examples such as *a new training regime*, *a regime of physical therapy*, *a regulatory regime for data privacy*, *a dietary regime*, or *the regime of international law*. In these cases, *regime* straightforwardly denotes a systematic plan or set of rules, with no inherent negative evaluation. The prosody of *regime* is therefore more accurately described as system-control, with its association with negatively evaluated governments being a pragmatic consequence of its meaning intersecting with a common domain of discussion.

Diachronic Evidence: Stability and Change

While the synchronic analyses above demonstrate that semantic profiles better explain current distributional patterns than absorbed prosodies, diachronic evidence provides crucial additional support for the profile-based account. If semantic prosodies were truly absorbed from repeated collocational exposure, we would expect to see gradual semantic change as words progressively take on evaluative coloring from their contexts. Conversely, if profiles are primary and relatively stable, we should find that core semantic features persist across centuries even as specific collocational patterns shift with changing discourse priorities and lexical competition.

Historical corpus evidence supports the profile stability hypothesis. Examination of the Oxford English Dictionary's historical citations and published diachronic corpus studies reveals that the semantic profiles identified above have remained remarkably consistent across long periods, even as the specific words with which they collocate have changed.

Consider *cause*, the most extensively documented case. The OED records transitive causative uses of *cause* from the fourteenth century onward, with the core meaning of *bring about, produce, or occasion* remaining stable. Middle English citations show *cause* already appearing with both negative outcomes and neutral processes. A 1390 citation has *cause grete harme*, while a 1485 text uses *cause joye and plesaunce*. Early Modern English texts of the sixteenth and seventeenth centuries show similar flexibility: Shakespeare employs *cause* with both *cause of grief* and *cause of joy*. The agentive, force-dynamic profile—requiring a salient agent directly bringing about an effect—is evident across all historical periods.

What changes over time is not the profile but the specific nominal collocates, which reflect the concerns and discourse priorities of different eras. Medieval texts show *cause* with *death*, *harm*, *woe*, and *grief* (reflecting medieval preoccupations with mortality and suffering), but also with *prosperity*, *increase*, and *multiplication* (in agricultural and religious contexts). Early Modern texts add collocates like *disturbance*, *confusion*, and *tumult* (reflecting political instability), while retaining positive uses in expressions like *cause for thanksgiving*. Modern corpora show *cause* with characteristically contemporary problems: *pollution*, *unemployment*, *stress*, *climate change*. Yet across this six-hundred-year span, the verb's agentive profile—requiring conceptualization of forceful, direct causation—remains constant.

Critically, counterexamples to the negative skew persist throughout the historical record. Each period contains documented uses of *cause* with positive or neutral outcomes, demonstrating that the profile has always licensed such uses even when they are statistically less frequent. A strong prosodic account would predict that as negative associations accumulated over centuries, positive uses should become increasingly marked or eventually impossible. Instead, we find stable flexibility: the proportion of negative to positive uses may vary somewhat across periods (depending on what types of events are topical), but both remain possible because both are licensed by the underlying profile.

This diachronic stability extends to the other words examined. The noun *regime*, borrowed from French *régime* in the late eighteenth century with the core meaning *system of government or rule*, initially appeared in relatively neutral political contexts. The OED's earliest citations (1792 onwards) use *regime* to describe various governmental systems without inherent negative evaluation: *the ancient regime of France*, *the present regime*, *under the regime of Louis XIV*. The system-control profile is evident from the beginning—*regime* denotes formalized, systematic administration.

The increasing association with authoritarian or negatively-evaluated governments emerges primarily in twentieth-century usage, particularly in reference to fascist, communist, and military dictatorships. This shift reflects changing geopolitical realities and discourse patterns rather than semantic change in the word itself. Crucially, non-political uses persist throughout: medical and scientific texts of the nineteenth and twentieth centuries routinely employ *regime* to describe *treatment regimes*, *dietary regimes*, and *exercise regimes*. The system-control profile remains stable and continues to license uses across political and non-political domains, even as the statistical distribution shifts toward political contexts due to their salience in news media and academic writing.

This diachronic pattern—stable profiles, shifting collocational frequencies—supports the theoretical claim that profiles are primary. The observed distributional patterns at any given moment reflect the interaction between stable semantic constraints and historically contingent discourse priorities. When discourse concerns shift (e.g., new types of problems become salient, different political systems dominate global attention), the specific collocates change, but the profile remains constant, continuing to constrain which contexts are felicitous.

This finding also clarifies the relationship between usage-based and profile-based accounts. Over very long timescales (centuries), repeated usage patterns can indeed lead to semantic change through grammaticalization and conventionalization processes (Traugott & Dasher, 2002; Bybee, 2010). The profiles analyzed here are themselves historical products. However, at any given synchronic moment, these relatively stable profiles function as constraints that shape rather than result from distributional patterns. The relationship is thus more nuanced than simple unidirectional causation: diachronically, usage can shape meaning, but synchronically, meaning shapes usage. The temporal scale matters enormously to the causal story.

Reconceptualizing Directionality

The case studies and diachronic evidence presented challenge the simplistic narrative of semantic prosody as usage-based absorption of evaluative coloring. Instead, they collectively support a model where a word's inherent semantic profile acts as the primary filter, attracting it to congruent contexts within human discourse. This lexical-semantic-first approach directly resolves several theoretical impasses while pointing toward a more integrated understanding that acknowledges both stability and dynamism in lexical meaning.

First, this model provides principled explanation where the prosody model offered only description. Documenting that *cause* frequently appears with negative outcomes is an important empirical finding, but it does not explain why. The analysis demonstrates that because *cause* encodes an agentive profile, and negatively evaluated events are more frequently construed as requiring a salient, forceful agent, the distributional pattern follows as a natural consequence. Similarly, the pattern for *regime* is explained by its system-control profile intersecting with the discursive prominence of authoritarian governments. The explanation moves from stable semantic features to observed distribution, reversing the prosodic model's direction of explanation.

Second, it elegantly accounts for the introspection problem. Native speakers are intuitively aware of core semantic profiles—they know *cause* implies directness, *utterly* means completeness, and *regime* implies a system. What they are not consciously aware of is the pragmatic skew that arises when those profiles interact with asymmetries in what we talk about. The negative tendency of *cause* is a byproduct of discourse priorities, not a component of its meaning, which explains why introspection into meaning fails to reveal it. Corpus linguistics uncovers the skew; lexical-semantic analysis explains its origin.

Third, the model seamlessly integrates counterexamples, which pose persistent problems for strong prosody claims. For every word examined, atypical uses in neutral or positive contexts are shown to be fully licensed and comprehensible through the same core semantic profile that explains the dominant pattern. The markedness or rhetorical effect of expressions like *cause for celebration* or *joy set in* stems from tension between the word's profile and a less typical context, not from violating a connotative rule.

However, this reconceptualization should not be misunderstood as rejecting usage-based insights entirely. Rather, it argues for theoretical integration that acknowledges different timescales and mechanisms of semantic change. Over diachronic timescales, usage patterns undoubtedly shape lexical representations through processes of grammaticalization, semantic change, and the conventionalization of pragmatic inference (Traugott & Dasher, 2002). The semantic profiles analyzed here are themselves historical products of language use. The critical point is that synchronically—at the level where corpus linguists observe patterns and speakers produce utterances—these profiles function as relatively stable constraints that guide rather than result from collocational patterns.

This suggests a more nuanced model of directionality that recognizes multiple levels of organization. At the diachronic level, repeated usage patterns can lead to semantic change, with pragmatic inferences becoming conventionalized over generations of speakers (Bybee, 2010). At the synchronic level, however, established semantic profiles constrain and predict usage patterns. The relationship is best understood not as unidirectional but as involving different types of causality operating at different timescales. Corpus patterns at a given historical moment reflect the interaction between inherited semantic architecture and current discourse pressures, rather than being purely emergent from those pressures.

Statistical Learning of Semantic Structure

The parallel success of large language models in acquiring human-like semantic competence from purely distributional training provides converging evidence for the profile-based account, while also revealing a structural limitation of distributional learning that the profile-based account predicts and the absorption account does not. Three observations are particularly telling regarding how statistical patterns encode semantic structure.

First, models demonstrate systematic generalization to counterexamples. GPT-4, Claude, and similar systems readily produce expressions like *foster resentment* and *cause celebration* without requiring specific training on these collocations (OpenAI, 2023; Anthropic, 2024). This suggests they have learned something more abstract than collocation frequencies—something resembling the facilitative and agentive profiles respectively. If corpus statistics encoded only surface associations between words and evaluative charges, we would expect models to resist producing low-frequency combinations that violate dominant patterns. Instead, models flexibly deploy words according to constraints that permit but do not require particular evaluative contexts. This flexibility indicates that distributional patterns carry information about underlying semantic architecture.

Recent probing studies have confirmed that language models develop representations sensitive to semantic features like agentivity, causality, and aspectual properties (Ettinger, 2020; Weir et al., 2020). These features emerge from distributional training without explicit supervision, suggesting they are encoded in the statistical patterns themselves. When *cause* appears with forceful impositions across diverse syntactic frames and discourse contexts, the consistent pattern encodes information about agentivity that statistical learning mechanisms can extract.

Second, language models show gradient acceptability judgments that align with profile-context fit rather than simple frequency (Lau et al., 2020). When prompted to rate the naturalness of expressions, models demonstrate sensitivity to the interaction between semantic profiles and conceptual appropriateness. *Joy set in* receives lower ratings than *peace set in* not because the model has learned that *set in* is negative, but because the distributional evidence encodes aspectual constraints: gradual-onset semantics fit better with states like peace (which can be conceived as gradually establishing) than with emotions like joy (typically construed as more immediate). Similarly, models rate *utterly wonderful* as slightly marked compared to *utterly terrible*, reflecting the pragmatic asymmetry in intensification conventions rather than a rigid negative prosody.

Third, models exhibit cross-construction and cross-domain transfer that reveals learning of abstract semantic features. Having encountered *regime* predominantly in political-historical contexts during training, models appropriately extend it to *exercise regime*, *dietary regime*, and *drug regime* in medical contexts (Misra et al., 2020). This transfer is predicted by the system-control profile but would be difficult to explain if the model had simply learned a context-specific negative association. The statistical signature of system-control—appearing in contexts describing structured, systematic administration regardless of domain—provides sufficient information for the model to extract the generalizable constraint.

These observations suggest that corpus statistics are informationally rich in ways that strong prosodic accounts underappreciate. Distributional patterns encode not merely which words co-occur but structural information about why they co-occur—information about the semantic features that make certain combinations conceptually appropriate. When *cause* appears with forceful impositions, when *foster* appears with gradual developments, when *regime* appears with systematic structures, the distributional evidence carries information about agentivity, facilitation, and system-control respectively. Statistical learning mechanisms in both human cognition and artificial neural networks can extract these structural regularities, enabling generalization beyond surface frequencies.

The Limits of Distributional Learning

The same evidence that supports profile-based interpretation of language model success also reveals a structural ceiling on what statistical learning of any scale can achieve. This limitation deserves explicit articulation because it distinguishes the profile-based account from accounts that would expect distributional methods, scaled sufficiently, to approximate semantic competence arbitrarily closely.

Contemporary language models are not pure-statistics systems. They are aggressive compressions of distributional information into a small parameter set, with the compression introducing distortions—lossy approximations, training-induced biases, dimensionality reductions that lose fine-grained co-occurrence structure. Many of the observable failures of current systems—hallucination, inconsistency, semantic imprecision in low-frequency domains—are traceable to compression artifacts rather than to limitations of distributional information itself. A system that operated on complete distributional information without compression—a theoretical full-statistics system that cannot in fact be built at scale—would perform substantially better than current implementations across most of the semantic landscape. The compression-induced failures would simply disappear.

However, performing better is not the same as performing well enough for semantic precision. The profile-based account predicts that even a full-statistics system would face a structural ceiling. Distributional information, however completely tracked, measures co-occurrence: which forms appear together, with what frequency, in what contexts. What it cannot measure is the conventionalization structure that underlies the distribution—the fact that speakers have settled on a pairing, that the pairing is intersubjectively stable as a commitment, not merely statistically frequent. Two scenarios are distributionally indistinguishable but semantically distinct under this view. A word whose meaning is genuinely settled in the lexicon and a word that happens to appear in similar contexts at similar frequencies through accumulated discourse patterns produce identical distributional signatures. A full-statistics system has no resource to distinguish stability-of-commitment from stability-of-co-occurrence; both produce the same statistical trace.

This is the categorical gap that distinguishes the profile-based account from absorption accounts. The gap is not engineering. It is not the kind of gap that scaling addresses. The semantic precision that profiles afford rests on the intersubjective stabilization of form-meaning pairings within a speaker community—a fact about coordinated commitments rather than about token frequencies. Distributional patterns are the surface trace of those commitments; they are reliably correlated with the underlying structure but are not identical to it. A system that operates only on the trace can approximate the structure with high but not unlimited fidelity. The failure modes of current language models in the high-fixity range—formal reasoning, citation accuracy, mathematical content—are not artifacts of insufficient training but expressions of this categorical gap (Mozahid, 2026c).

What this implies for the directionality debate is that the strong absorption thesis is wrong even in its most charitable form. It is not the case that distributional patterns, if accumulated in sufficient quantity, would constitute the meaning of a word. They constitute a reliable observable correlate of the meaning, generated by speakers whose coordinated semantic commitments leave statistical traces in the corpus record. Statistical learning succeeds at extracting structure to the extent that the trace preserves it. Where the trace is informationally complete (the middle ranges of conventionalization, where profiles are systematic but not rigid), statistical learning approximates semantic competence closely. Where the trace is informationally degenerate with respect to the semantic distinctions that matter (the highest ranges of conventionalization, where meaning is definitionally fixed and the trace shows no variance that statistics can exploit), statistical learning ceases to be a faithful proxy. This pattern of capability and limit is precisely what the profile-based account predicts, and it is precisely what current and prospective distributional systems exhibit.

Explicit Predictions and Theoretical Contrasts

To clarify the empirical consequences of the profile-based account and distinguish it from competing theories, this section articulates explicit predictions that differentiate the two approaches. If semantic prosodies represent absorbed evaluative features, as strong usage-based accounts claim, several predictions follow. First, counterexamples to dominant patterns should sound anomalous, requiring special pragmatic licensing or ironic interpretation. Second, acceptability should correlate primarily with collocational frequency—expressions that violate statistical norms should receive consistently low naturalness ratings. Third, language models trained on distributional data should overgeneralize dominant patterns and resist producing statistically atypical combinations. Fourth, cross-construction and cross-domain transfer should be limited, as prosodies would be tied to specific collocational contexts. Fifth, distributional signatures of evaluative coloring should require large corpus sizes to consolidate, since they emerge through accumulated frequency rather than being already present in the lexical entry.

Conversely, if semantic profiles constitute primary constraints that shape usage, different predictions emerge. First, counterexamples should be comprehensible when the profile fits the context, even if statistically infrequent. Second, acceptability should be predicted by profile-context fit rather than raw frequency—expressions that match the profile should receive higher ratings even when uncommon. Third, language models should generalize systematically beyond frequency, producing both prototypical and counterexample uses when the profile licenses them. Fourth, cross-construction and cross-domain transfer should follow profile constraints, enabling systematic extension to novel contexts. Fifth, the distributional signatures of profiles should be scale-invariant—detectable at small as well as large corpus sizes—because the constraint generating the distribution is present in the lexical entry independent of how much corpus data has accumulated.

The evidence consistently supports the profile-based predictions. As documented in the case studies, counterexamples like *foster resentment*, *cause celebration*, and *peaceful routine set in* are fully comprehensible without requiring ironic readings. Acceptability patterns align with profile-context fit: *peace set in* sounds more natural than *joy set in* not due to frequency but because peace is more compatible with gradual-onset semantics. Language models demonstrate systematic generalization, appropriately producing both *policies that foster growth* and *policies that foster inequality* based on facilitative profile constraints rather than statistical dominance. Cross-domain transfer is profile-constrained: *regime* extends from political to medical contexts because both involve system-control, not through simple analogical extension. The fifth prediction—scale-invariance—remains to be empirically tested, but it is articulated precisely enough that its eventual confirmation or disconfirmation would constitute decisive evidence between the accounts.

These contrasting predictions provide a framework for future empirical research that could definitively adjudicate between the competing accounts. The following section outlines specific experimental designs that would test these predictions directly.

Experimental Validation: Proposed Studies

While the theoretical reanalysis presented here is grounded in established corpus patterns, supported by diachronic evidence, and consistent with language model behavior, direct experimental validation would strengthen the empirical foundation. This section proposes five experimental studies designed to test the core predictions of the profile-based account against those of strong prosodic theories.

Study 1 would employ acceptability judgment methodology to test whether profile-context fit predicts naturalness better than collocational frequency. Participants would rate the naturalness of expressions on seven-point Likert scales. Critical items would include profile-matching but low-frequency uses (e.g., *foster resentment*, *cause celebration*, *utterly beautiful*) compared to high-frequency uses (e.g., *foster growth*, *cause problems*, *utterly terrible*) and profile-mismatching controls. The profile account predicts that profile-matching items should receive high ratings regardless of frequency, while prosody accounts predict strong frequency effects. Additional items would test gradient effects: expressions that partially match profiles should receive intermediate ratings. For instance, *foster apathy* (gradual development of negative state) should rate higher than *foster rage* (rage is typically construed as sudden, not gradual). Statistical analysis using mixed-effects models would determine whether profile features (coded based on the analyses presented here) predict ratings beyond variance explained by frequency alone.

Study 2 would employ semantic priming to test whether activating profile features facilitates processing of appropriate target words. Participants would make lexical decisions on target words following prime stimuli that activate relevant semantic features. For the agentivity profile of *cause*, primes might be short vignettes describing forceful, direct imposition of outcomes (e.g., *the boulder crashed into the wall and...*). For the facilitative profile of *foster*, primes would describe gradual enablement (e.g., *over months, the supportive environment allowed...*). The profile account predicts that priming profile-relevant features should facilitate processing of the target word regardless of whether the outcome is positive or negative. If *cause* has absorbed negative prosody, priming positive forceful events should show interference rather than facilitation.

Study 3 would test cross-linguistic generality of profile-pattern relationships. If the patterns documented here reflect genuine semantic profiles rather than English-specific conventionalization, similar profiles should attract similar distributional patterns in typologically diverse languages. This study would replicate the corpus analyses for semantically parallel verbs in languages with different causative systems. For instance, Romance languages have multiple causative constructions with different force-dynamic profiles (Wolff, 2003); the prediction is that the most agentive causatives should show negative skews similar to *cause*, while facilitative causatives should pattern like *foster*. This would demonstrate that the relationship between profiles and distributions reflects conceptual structure rather than arbitrary English conventions.

Study 4 would employ computational probing methods to test whether language model representations encode the proposed semantic features. Recent work in interpretability has developed techniques for extracting and measuring semantic features in neural network representations (Hewitt & Manning, 2019; Tenney et al., 2019). This study would train probing classifiers to detect profile-relevant features (agentivity, facilitation, gradual-onset, etc.) in the contextualized embeddings that language models generate for target words. The profile account predicts that these features should be robustly extractable and should predict the model's generation behavior better than simple frequency statistics. Specifically, the strength of profile features in a given context should predict the model's probability of using the target word in that context, even controlling for how frequently the model encountered similar contexts during training.

Study 5 would test the scale-invariance prediction articulated in the methodological section. A freely available corpus (BNC, COCA samples, or the Open American National Corpus) would be partitioned at progressively decreasing scales—halves, quarters, eighths, sixteenths, thirty-seconds. For each scale, profile-relevant collocational distributions would be computed for a target set of items including the six words analyzed in this study and a comparison set of items selected to vary in profile fixity. Convergence of rankings and qualitative shape of distributions across scales would be quantified using rank-correlation measures. The profile account predicts substantial convergence even at modest scales, with the strongest convergence for high-fixity profile features and weaker convergence for low-fixity features. An absorption account predicts substantial reorganization of structure as scale increases, with no principled basis for distinguishing convergence patterns across feature types. A pilot version of this study could be conducted with thirty target items across four corpus sizes using standard concordance tools and would already provide informative evidence on the central prediction.

These five studies would provide converging evidence from acceptability judgments (testing conscious linguistic intuitions), priming (testing automatic processing), cross-linguistic comparison (testing universality), computational modeling (testing learning mechanisms), and scale analysis (testing the specific empirical signature that distinguishes profile-based from absorption-based accounts). Positive results would strongly support the claim that semantic profiles are psychologically real, cross-linguistically robust, learnable from distributional evidence, and detectable in distributional data at any scale because they are properties of the lexical entry rather than properties of accumulated frequency.

The Lexical-Encyclopedic Consequence

The directionality thesis defended in this paper has a consequence that warrants explicit registration even though its full development belongs elsewhere. If semantic profiles drive distributional patterns rather than being absorbed from them, and if the profiles in question—agentivity, facilitation, system-control, intensive completeness, volitional binding, gradual onset—integrate features that traditional accounts assign separately to linguistic meaning and to world knowledge, then the directionality thesis is not neutral with respect to the lexical-encyclopedic boundary. It actively destabilizes the categorical version of that boundary.

This is not a novel destabilization. Cognitive linguistic traditions from Langacker (1987) and Fillmore (1982) onward have argued that lexical meaning is inherently encyclopedic, that frame semantic structures spanning linguistic and world knowledge are required to understand even apparently simple lexical items, and that the rigorous partitioning of meaning into "lexical" and "encyclopedic" components is theoretically unmotivated. Fillmore's (1982) analysis of *bachelor* established the point clearly: appropriate application of even this paradigmatic case of definitional simplicity requires extensive cultural and social knowledge about marriage practices, social expectations, and typical life trajectories that cannot be cleanly excised from the word's meaning. Langacker's (1987) cognitive grammar generalizes the position into a wholesale rejection of the lex-enc dichotomy.

The semantic profiles examined in this paper inherit and extend that tradition. To take the central case: the profile of *cause* involves understanding force-dynamic relationships (arguably conceptual), typical causation scenarios in human experience (arguably encyclopedic), and grammatical consequences such as which nominals can serve as subjects and objects (arguably linguistic). Attempting to partition the profile into purely lexical versus encyclopedic components would be artificial and theoretically unmotivated. The profile functions as a unified whole in constraining usage, and this functional unity is what the present analysis requires.

Mounting evidence from neighboring domains converges on the same conclusion. Large language models acquire both grammatical competence and factual knowledge from identical distributional training without architectural separation corresponding to the lex-enc dichotomy, suggesting that the distinctions traditionally drawn between knowledge types are not as categorical as theory assumes. Research in language testing demonstrates the empirical inseparability of linguistic proficiency and domain knowledge in actual language use (Alderson & Urquhart, 1988; Clapham, 1996; Bachman & Palmer, 1996). I have developed the full theoretical articulation of this position in separate work (Mozahid, 2025, 2026a, 2026b), where the lex-enc distinction is reconceptualized as a gradient of conventionalization operationalized through the alpha coefficient. The present paper takes the gradient picture as a working assumption rather than as its central argumentative burden. What matters here is that the profile-based account does not require maintaining a categorical lex-enc boundary, and indeed coexists naturally with the gradient view defended in the related work.

Implications and Applications

Linguistic Theory

For linguistic theory, this reconceptualization demonstrates the necessity of integrating lexical-semantic analysis with corpus methodologies. As Hunston (2007, p. 53) cautions, corpus data are evidence of meaning rather than meaning itself. Corpora are unparalleled descriptive tools revealing hidden patterns, but those patterns demand explanation. A theory that stops at identifying a negative prosody provides a label, not an explanation. Explanatory power comes from linking distribution to semantic architecture. This approach also resolves tension between the stability of lexical meaning and fluidity of use by positing a stable core that governs, but does not rigidly determine, contextual deployment.

The framework developed here suggests productive directions for future research. First, explicit formalization of semantic profiles using feature geometry or frame semantic representations could make predictions more precise and testable. Rather than informal descriptions of agentive or facilitative meanings, researchers could specify feature bundles that predict both prototypical uses and acceptable extensions. The gradient conventionalization framework developed in Mozahid (2025, 2026a) offers one such formalization, operationalizing profile components through the alpha coefficient and providing precise mappings between feature fixity and entailment strength. Second, cross-linguistic investigation could test whether similar profile-pattern relationships hold in typologically diverse languages, as proposed in Study 3 above. Third, integration with psycholinguistic research on lexical representation and processing would test the psychological reality of profiles and determine how they interact with frequency effects in online comprehension and production.

Lexicography

For lexicography, the implications are practical and actionable. Dictionaries that note collocational tendencies of words like *cause* or *regime* provide valuable usage guidance. However, definitions should strive to capture underlying semantic profiles that motivate these tendencies. Defining *cause* with explicit reference to its direct agentivity, rather than merely noting its frequent use with something bad, provides a more generative and accurate account that explains both common negative uses and less frequent positive ones.

A revised entry for *cause* might read: *To bring about or produce (an effect or result) through direct, forceful action or influence; implies a salient agent or potent force directly effecting change. While evaluatively neutral, cause frequently appears with negative outcomes in discourse, as harmful events are often conceptualized as requiring forceful agents. Compare lead to (less direct causation), foster (facilitative causation), result in (consequence without implied agency)*.

Similarly, defining *regime* to emphasize its system-control semantics, with a usage note about frequency in political contexts, would better serve dictionary users than simply labeling it as often pejorative. This approach empowers users to understand not just what words tend to mean but why they pattern as they do, providing generative knowledge applicable to novel contexts.

Natural Language Processing

For natural language processing and computational linguistics, the directionality debate has direct practical consequences. Systems for sentiment analysis, machine translation, and text generation that rely on simplistic polarity lexicons often fail because they mistake pragmatic skew for semantic rule. A sentiment analyzer trained on collocational frequency of *cause* might incorrectly tag *cause for celebration* as negative. A more robust system would incorporate features representing semantic profiles (e.g., ±agentive, ±intensive, ±facilitative) extracted through the probing methods proposed in Study 4.

Understanding that *cause* encodes an agentive profile allows systems to better interpret why *cause economic growth* sounds more forceful and less natural than *foster economic growth*. The difference is not that *cause* is negative while *foster* is positive, but that forceful direct causation is pragmatically less felicitous for describing gradual economic processes than facilitative causation. NLP systems that model these semantic distinctions achieve more nuanced, human-like language understanding (Mohammad & Turney, 2013; Ribeiro et al., 2020).

Recent advances in neural language models suggest that these systems implicitly learn approximations of semantic profiles from distributional training, as their successful generalization demonstrates. However, the structural ceiling on distributional learning discussed above implies that explicit modeling of profiles—rather than reliance on increasingly large statistical approximations of them—will be necessary for systems that aspire to operate reliably across the full range of semantic content. Architectures that learn structured semantic representations, perhaps through techniques like probing classifiers that identify profile features in learned embeddings or through hybrid systems that combine neural learning with symbolic semantic features, could achieve better zero-shot generalization and more robust performance on edge cases. Such approaches would combine the statistical learning capacity of neural networks with the structured knowledge traditionally associated with symbolic AI (Bender & Koller, 2020; Marcus, 2020). The relationship between distributional learning capacity and the conventionalization gradient is developed in more detail in Mozahid (2026c).

Language Pedagogy

For language pedagogy, this perspective demonstrates the value of teaching meaning in depth rather than presenting students with lists of words carrying bad or good associations.

Instructors can explain core semantic features that make particular associations likely. Teaching that *commit* involves binding choice helps learners understand its range from *commit a crime* to *commit to a partner*. This deeper understanding is more transferable and empowering than memorizing collocational lists.

Consider pedagogical approaches to *cause* and *foster*. Traditional instruction might present them as near-synonyms differing primarily in connotation: *cause* is used with bad things, *foster* with good things. This would confuse learners when they encounter *cause excitement* or *foster dependence*. A profile-based approach instead explains: *cause* implies direct, forceful bringing about; *foster* implies creating conditions that enable gradual development. In English discourse, we more often frame harmful events as forcefully imposed and beneficial developments as gradually nurtured, which is why you'll see these patterns—but the words themselves are evaluatively neutral. This explanation prepares learners for both prototypical and creative uses.

Language textbooks could benefit from exercises that help learners internalize semantic profiles through metalinguistic reflection. Rather than gap-fill exercises focusing on correct collocations, activities could ask learners to explain why certain combinations sound natural or marked: *why does* peace set in *sound better than* joy set in? *Consider what* set in *means about how a state begins*. Such reflection develops deeper semantic understanding that supports flexible, native-like use.

Conclusion

This paper has demonstrated a significant reconceptualization in how linguists understand the phenomenon often labeled semantic prosody. Through detailed analysis of *cause*, *utterly*, *commit*, *set in*, *foster*, and *regime*, it has shown that their striking collocational patterns are better understood as consequences rather than causes of their connotative meanings. The direction of explanation flows from stable lexical-semantic profiles to observed usage patterns, mediated by discourse pragmatics, rather than from usage to meaning as the strong prosodic account suggests.

The negative skew of *cause* derives from its agentive profile interacting with the types of events we most frequently report; the negative tendency of *utterly* stems from its intensive meaning meeting conventions of evaluative discourse; the association of *commit* with transgression arises from its volitional-binding core attracting it to contexts of weighty, deliberate acts; the link between *set in* and undesirability is rooted in its semantics of gradual onset matching the temporal structure of commonly discussed negative processes; the positive lean of *foster* emerges from its facilitative nature aligning with how we describe cultivated outcomes; and the negative aura of *regime* is a consequence of its system-control profile intersecting with dominant political discourse topics.

Corpus-attested counterexamples for each word undermine claims of inherent, fixed evaluative prosodies and confirm that core semantic profiles remain constant, generative sources of meaning across diverse contexts. Diachronic evidence from historical corpora and the Oxford English Dictionary demonstrates that these profiles have remained stable across centuries of English usage, even as specific collocational patterns have shifted with changing discourse priorities. This historical stability of semantic architecture, concurrent with variability in distributional patterns, directly supports the claim that profiles drive usage rather than absorbing evaluative charge from it.

The framework deployed here avoids the circularity that would otherwise undermine the reanalysis. Profile characterizations are derived from lexicographic evidence—principally the Oxford English Dictionary's historical entries and theoretical work on causation, force dynamics, aspect, and intensification—rather than from the synchronic corpus distributions they are then used to explain. The independence of profile evidence from distributional evidence is what allows the analysis to function as prediction and confirmation rather than as observation relabeled. A further empirical prediction—that profile-driven distributions should be scale-invariant across corpus sizes—has been articulated explicitly, with the test described in operationally specific terms. The prediction is offered as a theoretical commitment of the framework, to be confirmed or disconfirmed empirically when computational means become available. A framework that generates falsifiable predictions before it can run them is doing the work that theoretical frameworks are supposed to do.

Moving beyond semantic prosody means moving beyond description to explanation. It requires recognizing that words carry inherent semantic shapes—structured sets of features defining prototypical characteristics—that selectively attract them to particular corners of conceptual and discursive space. The connotative patterns observed in corpora are results of this attraction, filtered through the preoccupations and priorities of human discourse.

This lexical-semantic-first model does not deny the influence of usage on language change over long timescales, but it establishes the explanatory priority of encoded meaning for understanding stable, systematic connotative behaviors of the core lexicon at any given synchronic moment. By shifting from a model of connotative absorption from context to one of profile-constrained deployment according to inherent features, we can reintegrate sophisticated understanding of connotation into the heart of linguistic theory.

The reconceptualization defended here has a consequence for the traditional lexical-encyclopedic boundary that should be registered explicitly. The semantic profiles examined—agentive, intensive, facilitative, gradual-onset, volitionally binding, system-control—inherently integrate features that traditional accounts would partition across this putative divide. Evidence from large language models, which acquire both linguistic competence and world knowledge from identical distributional patterns, demonstrates the permeability of this boundary in practice.

The framework advocated here does not claim complete dissolution of all distinctions but is committed to the view that the boundary is gradient, context-dependent, and permeable rather than categorical and fixed. The full theoretical articulation of this commitment, including its formalization as a gradient of conventionalization operationalized through the alpha coefficient, is developed in separate work (Mozahid, 2025, 2026a, 2026b).

The proposed integrated model acknowledges that lexical representations are shaped by usage diachronically while maintaining that synchronic distributional patterns are better explained by stable semantic architecture interacting with discourse pragmatics. This reconceptualization bridges rich corpus data with explanatory depth from formal and cognitive semantics, offering a more complete and principled account of how words mean and how that meaning shapes the way we deploy them to understand, describe, and evaluate our world.

The parallel implications for understanding statistical language models add contemporary urgency to these theoretical debates. The success of systems like GPT-4 and Claude in acquiring human-like semantic competence from purely distributional training is not mysterious once we recognize that corpus statistics encode information about semantic structure. Collocational patterns are not arbitrary associations but consequences of semantic profiles interacting with discourse pragmatics. Statistical learning mechanisms, whether in biological or artificial neural networks, can extract these structural regularities precisely because they are systematically encoded in distributional patterns. At the same time, the structural ceiling on distributional learning that the profile-based account predicts is itself visible in the observable failure modes of current systems. Hypothetical full-statistics systems, operating on complete distributional information without compression, would perform substantially better than current implementations across most domains, removing the failures that compression artifacts introduce. But they would not achieve semantic precision through scale, because the gap between distribution and meaning is not engineering but categorical. Distribution measures co-occurrence; meaning resides in the intersubjective stabilization of form-meaning pairings within a speaker community. Distribution is the reliable trace of meaning, not meaning itself. The opposition between mere statistics and real understanding need not be drawn as a strict dichotomy, but neither should it be collapsed: the trace is informationally rich, recoverable to a substantial degree by statistical learning, and yet not identical to what generates it.

Future research should pursue the experimental validation proposed here through psycholinguistic methods, cross-linguistic comparison to test the universality of profile-pattern relationships, corpus-scale analyses of the scale-invariance prediction, and computational modeling that operationalizes profile features for natural language processing applications. Investigating whether languages with different discourse conventions show similar or different mappings between profiles and distributions would illuminate the relative contributions of universal semantic structure versus language-specific pragmatic conventionalization. Developing explicit formal representations of semantic profiles, building on the gradient conventionalization framework, would enable more precise predictions and computational implementations.

By recognizing that semantic prosody research has discovered not evaluative absorption but the distributional fingerprints of semantic architecture, we achieve theoretical unification: corpus patterns become explicable through lexical semantics, usage-based and formal approaches are integrated across timescales, counterexamples are seamlessly accommodated, and the success and the limits of statistical language models in approximating human semantic competence both become understandable within a single account. The path forward requires continued dialogue between corpus linguistics, formal semantics, psycholinguistics, and computational approaches—each contributing essential insights to our understanding of how meaning shapes, and is shaped by, the statistical fabric of language use.

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